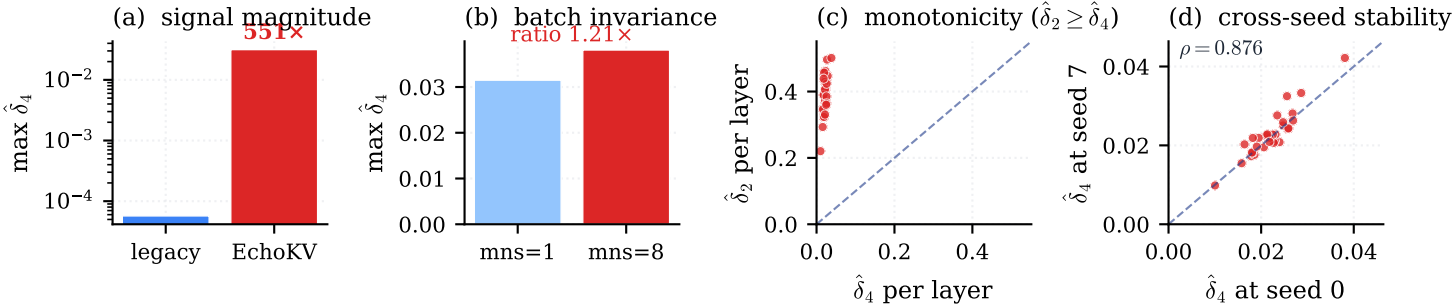
(a) signal magnitude
551×
max $\hat{\delta}_4$
$10^{-2}$
$10^{-3}$
$10^{-4}$
legacy
EchoKV
(b) batch invariance
ratio 1.21×
max $\hat{\delta}_4$
0.03
0.02
0.01
0.00
mns=1
mns=8
(c) monotonicity ($\hat{\delta}_2 \geq \hat{\delta}_4$)
$\hat{\delta}_2$ per layer
0.4
0.2
0.0
0.0
0.2
0.4
$\hat{\delta}_4$ per layer
(d) cross-seed stability
$\rho = 0.876$
$\hat{\delta}_4$ at seed 7
0.04
0.02
0.00
0.00
0.02
0.04
$\hat{\delta}_4$ at seed 0